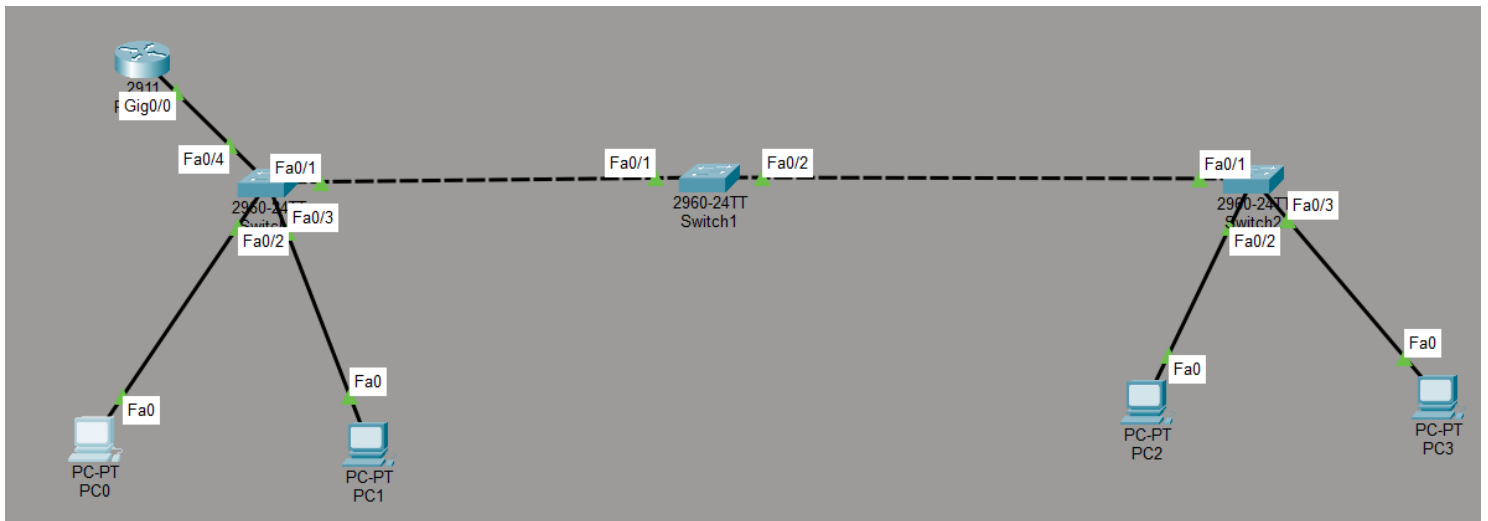


Q- Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.



Switch0 Configuration

The router is connected to Switch0 on **Fa0/4**. This port must be configured as a trunk.

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int f 0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
```

Router0 Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g 0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#int g 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Verification

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0    unassigned      YES unset    up          up
GigabitEthernet0/0.10 192.168.10.1    YES manual    up          up
GigabitEthernet0/0.20 192.168.20.1    YES manual    up          up
GigabitEthernet0/1    unassigned      YES unset    administratively down down
GigabitEthernet0/2    unassigned      YES unset    administratively down down
Vlan1           unassigned      YES unset    administratively down down
Router#
```

Ping Test Connectivity

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time=1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Inter-VLAN routing was successfully configured using the Router-on-a-Stick method. VLAN 10 (Sales) and VLAN 20 (Support) were assigned separate sub-interfaces on the router with appropriate IP addresses. The link between the router and Switch0 was configured as a trunk, allowing VLAN 10 and VLAN 20 traffic to reach the router.